PROJECT AND TEAM INFORMATION

Project Title

Quiz Application

Student / Team Information

Team Name: ***Team Id:***	*Bug Hunters* *T092*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	Bisht, Smriti – 240221598 240221598@geu.ac.in
Team member 2 *(Last Name, name: student ID: email, picture):*	Kumar Rai , Tejas – 24021935 24021935@geu.ac.in

Team member 3 *(Last Name, name: student ID: email, picture):*	Chamoli, Pranjal – 24021077 24021077@geu.ac.in
Team member 4 *(Last Name, name: student ID: email, picture):*	Semwal, Pranjal – 24021273 24021273@geu.ac.in

PROPOSAL DESCRIPTION (10 pts)

Students generally after studying a topic move on to the next one and the process just keeps on going, though it sounds seamless, the problem arises when the same topic is revisited, the student may find themselves unable to answer the questions based on the said topic, this is because their skills were not assessed after learning. Our project aims to solve this problem, it is a desktop-based java application that allows users to register, log in, attempt quizzes and view results while the administrators can manage questions and users. It supports multiple quiz categories and difficulty levels while using a timed mechanism and stores data persistently in a database. It also provides functionality of importing and exporting files. Though it sounds simple, the project incorporates most of the concepts in object-oriented programming, uses Java Swing for the graphical user interface, JDBC for database connectivity to ensure reliability and efficiency in performance.

Motivation

Manual quiz systems are slow, inefficient and prone to human error, this is where the automatic quiz system comes in picture though all the drawbacks of the manual systems are fixed, these systems are generally designed for large-scale platforms, making them unsuitable for small institutions, classrooms or individual learners. Our project aims to fill the gaps between the two. There is still a growing need for a lightweight, user-friendly quiz application that allows easy quiz creation, real time assessment immediate feedback moreover also have customisation options such that educators can tailor quizzes according to specific topics, difficulty levels or learning objectives.

State of the Art/ Current Solution

Today online quiz platforms are used by institutes to conduct assessments and evaluate performance automatically, these platforms are feature rich and support multiple choice questions, timers, automated grading and result storage in databases. Some platforms also provide analytic dashboards for teachers to monitor student performance. however, many of the such solutions are subscription based, complex to configure, or require continuous internet connectivity. As a result, there is a need for a customisable, standalone quiz application that provides essential assessment features with ease of use and flexibility.

Project Goals and Milestones

Goals:

- *T*o develop a Java-based quiz application with an interactive and user-friendly graphical interface.
- To implement secure user authentication and enable users to attempt quizzes with real-time evaluation and instant results.
- To provide persistent data storage using database connectivity for managing users, questions, and performance records.
- To create anefficient system for administrators to manage question banks and monitor user performance.

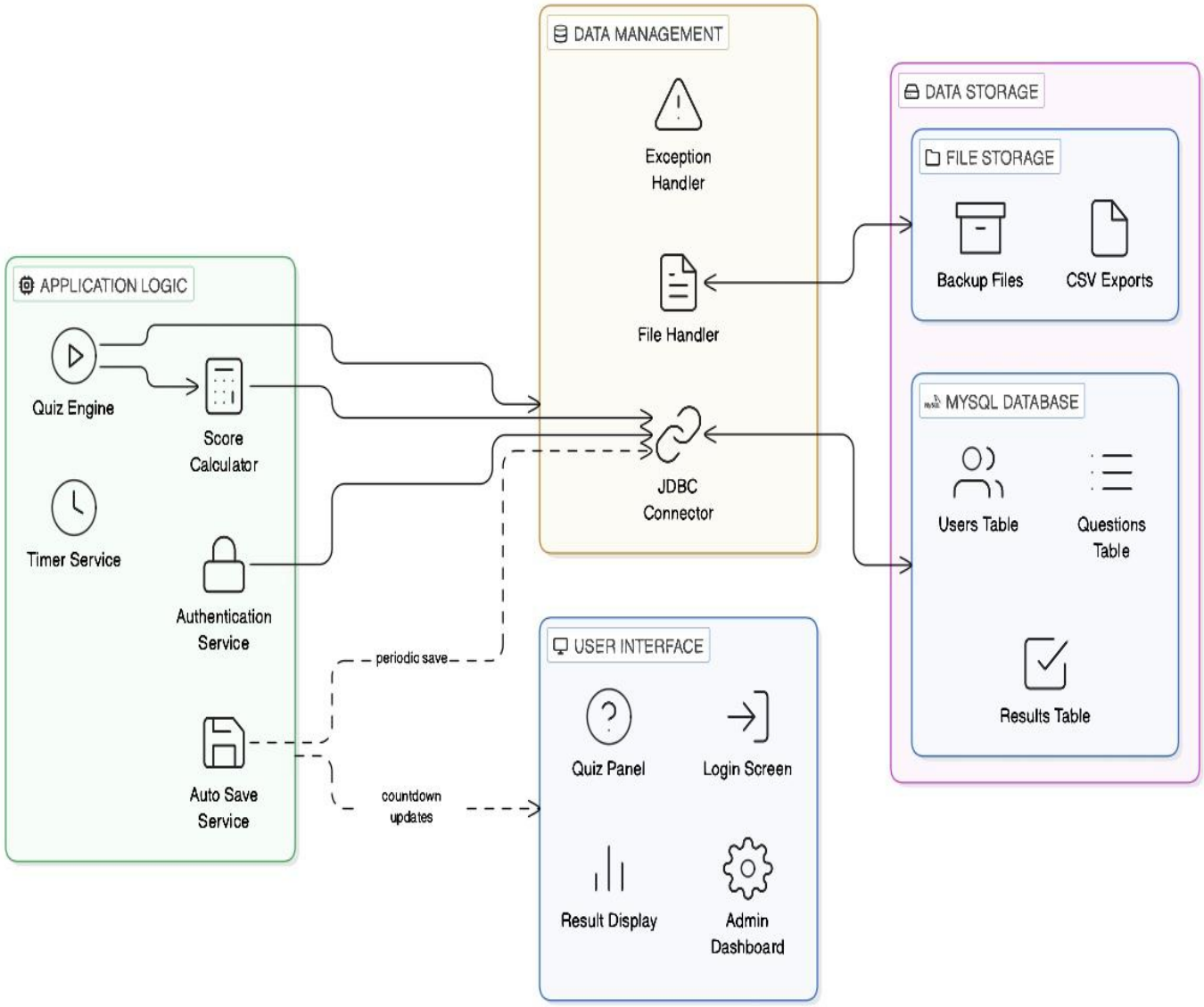
Milestones:

1. Identifying system requirements, user roles and core entities.
2. Design class structure, GUI flow and database schema.
3. Develop quiz engine, question navigation, answer validation and score calculation.
4. Implementation of import and export features and custom exception handling.
5. Improve interactivity by building screen for login, quiz interface, results and admin panel.
6. Establish JDBC connectivity and implementing CRUD operations.
7. Addition of advanced features such as timer functionality, multithreading and auto-save mechanism.
8. Performing unit and usability testing and finalisation of documentation.

Project Approach

The application will be developed as a Java desktop application using Object-Oriented Programming principles for modular and maintainable design. A layered architecture will separate the GUI, core logic, and data management components. Java Swing will be used to create an interactive user interface, including login, quiz, result, and admin panels. JDBC will enable database connectivity (MySQL/SQLite) for storing users, questions, and results. Core quiz functions such as question navigation, answer validation, and score calculation will be implemented using Java Collections. File handling will support import/export of quiz data, while exception handling will ensure reliable execution. Multithreading will be used for timers and real-time features without affecting GUI responsiveness. Testing will be performed to ensure accuracy, usability, and system performance.

System Architecture



Java Concepts Used in Each Module

APPLICATION LOGIC MODULE

Components: Quiz Engine, Score Calculator, Timer Service, Authentication Service, Auto Save Service

Java Concepts Used:

• Object-oriented programming is used to design classes such as QuizEngine , ScoreCalculator , TimerService that lay down the fundamentals of our project.

• Encapsulation protects sensitive data such as user credentials and scores by restricting direct access and unauthorised modifications.

• Inheritance is used to create user types such as administrators and students, allowing reusability of code.

• Multithreading supports the timer service by running countdown operations concurrently with quiz execution without interrupting user interaction.

• Exception handling ensures the system takes into consideration invalid inputs, login failures, or unexpected runtime errors.
• The collections framework is used to manage groups of questions, answers, and user responses .
• Serialization enables the auto-save feature by storing session data so that quiz progress can be restored later.

DATA MANAGEMENT MODULE
Components: File Handler, JDBC Connector, Exception Handler
Java Concepts Used:
• Java Database Connectivity enables communication between the application and the database.
• File input and output operations support reading backup files and writing exported quiz results.
• Custom exception handling such as DatabaseException and FileStorageException helps in managing database and file related errors.
• The try-with-resources mechanism ensures proper closing of database connections and file streams resulting in the prevention of leaking of resources.
• Logging mechanisms in the utility package help track system activity and errors, assisting in debugging and maintenance.

DATA STORAGE MODULE
Components: MySQL Database, File Storage
Java Concepts Used:
• Structured Query Language is used with Java to perform operations such as inserting, updating, retrieving, and deleting records related to users, questions, and results.
• JDBC drivers enable secure and efficient interaction with the MySQL database.
• File handling techniques support storage of backup files and exporting quiz results in CSV format.
• Connection pooling may be used to improve performance by reusing active database connections instead of creating new ones repeatedly.

USER INTERFACE MODULE
Components: Quiz Panel, Login Screen, Result Display, Admin Dashboard
Java Concepts Used:
• Java Swing or JavaFX is used to build the graphical user interface .
• Event handling mechanisms detect and respond to user actions such as button clicks and form submissions.
• The Model-View-Controller architecture acts as a separator for the user interface , logic and data .
• Layout managers are used to organise interface elements.

CROSS-MODULE CONCEPTS
Java Concepts Used Across the System:
• Packages are used to organize the application into logical modules such as user interface, services, data access, and models.
• Interfaces define common behaviors for services, enabling flexibility and reusability.
• Generics provide type safety when managing collections of questions, users, and results.
• Enumerations represent fixed sets of constants such as user roles and difficulty levels.
• Design patterns such as the Data Access Object pattern standardize database interactions, while the Singleton pattern ensures a single shared database connection instance.

Project Outcome / Deliverables

- A fully functional Java-based Quiz Application with an interactive graphical user interface.
- Secure user authentication system with role-based access for students and administrators.
- Quiz module with automated evaluation, instant results, and performance tracking.
- Database integration for regular storage of users, questions, and quiz results.
- Administrative tools for managing question banks and maintaining quiz content.

Assumptions

- The system will be used in a controlled environment with valid user credentials.
- The database server (MySQL/SQLite) is properly configured and accessible to the application.
- Quiz data such as questions and categories is preloaded by the administrator.
- Java Runtime Environment is installed in the System.

References

- Oracle Corporation. *Java Documentation*. https://docs.oracle.com/en/java/
- Oracle Corporation. *The Java™ Tutorials – Swing (GUI Development)*. https://docs.oracle.com/javase/tutorial/uiswing/
- Oracle Corporation. *JDBC Basics Tutorial*. https://docs.oracle.com/javase/tutorial/jdbc/
- MySQL. *MySQL Documentation*. https://dev.mysql.com/doc/
- SQLite. *SQLite Documentation*. https://www.sqlite.org/docs.html
- GeeksforGeeks. *Java Programming Tutorials*. https://www.geeksforgeeks.org/java/